THEJEESH G

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LeetCode: 301/3991 Solved (Easy 129 | Medium 142 | Hard 30)

EDUCATION

Bachelor of Technology in Computer Science Engineering *June 2023 – Present*
Vellore Institute of Technology (VIT), Chennai

TECHNICAL SKILLS

Programming Languages: Python, Java, C, C++, JavaScript
Frameworks & Libraries: React, Three.js, Node.js/Express, Bootstrap, NumPy, Pandas, Tailwind CSS
Tools & Platforms: Git, GitHub, AWS, MySQL, Neo4j, FAISS, Streamlit, Figma, Postman, Vercel
Core Concepts: Data Structures & Algorithms, Machine Learning, REST APIs, Object-Oriented Programming, System Design

WORK EXPERIENCE

Research Intern | VIT Chennai *May 2025 – July 2025*

- Built a Retrieval-Augmented Generation (RAG) system to evaluate and provide feedback on student answers, grounding an LLM on curated academic textbooks to ensure subject-accurate responses.
- Implemented a FAISS-based vector similarity search pipeline and a vector database for efficient storage and retrieval of embedded textbook content, enabling low-latency context lookup at inference time.
- Generated up to 50 unique response variations per session through iterative prompt engineering, improving the relevance and diversity of AI-generated feedback.
- Designed feedback validation and refinement workflows to continuously improve response accuracy against academic standards.
- Built the system on a modular, scalable architecture to support future extension across subjects and question types.

PROJECTS

DressFit – 3D Virtual Fitting Application *React, Three.js, react-three-fiber*

- Built a 3D virtual dress-fitting web application enabling real-time avatar rendering, scaling, and rotation in the browser.
- Implemented dynamic avatar scaling based on user body measurements (height, bust, waist, hips) to accurately reflect body shape.
- Designed dress customization logic with size and color/pattern selection, using geometric matching to determine fit accuracy.
- Currently extending the project with facial/body customization via Ready Player Me avatar integration.

Enterprise Knowledge Graph – Student Career Intelligence Platform *React, Node.js/Express, Neo4j*

- Built a graph-based platform modeling students, skills, courses, projects, and career roles in Neo4j using multi-hop relationships (HAS_SKILL, REQUIRES_SKILL, COVERS_SKILL).
- Designed backend graph traversal logic to compute career readiness scores, skill-gap analysis, and bridge-to-role recommendations.
- Implemented a counterfactual simulation engine modeling how new skills change readiness and unlock additional career paths.
- Built an explainable learning path finder ranking multiple recommendation strategies (fast-track, balanced, portfolio-first) with reasoning behind each.
- Integrated role-based access control (admin/student) via Express middleware for secure, scoped data access.

F1 Race Outcome Prediction *Python, Machine Learning, Jolpica Ergast API*

- Built a machine learning pipeline to predict Formula 1 race outcomes using historical race data from 2018 to the present, sourced via the Jolpica Ergast API.
- Trained the prediction model on 7 seasons of historical race data, accurately predicting the outcome of the Austria Grand Prix in real-world validation.
- Engineered a Python-based data scraping and preprocessing pipeline to collect and clean multi-season race, driver, and constructor data.
- Applied feature engineering and trained predictive models to forecast race results based on historical performance trends.
- Systematically debugged and resolved a recurring pipeline crash traced to cloud-sync interference during file writes, improving pipeline reliability across multi-season data runs.

Adaptive Video Streaming Framework *Java, FFmpeg, Multi-threading*

- Engineered an adaptive video streaming framework using a hierarchical client-server-relay architecture, sustaining playback at ~40 FPS.
- Implemented TCP-AIMD-based congestion control and selective retransmission protocols for reliable packet recovery.
- Used FFmpeg for real-time frame-level transcoding, optimizing bitrate adaptation and reducing latency.
- Architected a multi-threaded session management system to support concurrent client scalability, with LRU-based caching at intermediary nodes to improve throughput.

PUBLICATIONS

- Co-authored "Design and Evaluation of a Privacy-Preserving Blockchain Framework for Transparent Humanitarian Aid Distribution," accepted and presented at an IEEE conference, Madhya Pradesh, April 2026. Proposed SolidusAidFlow, a blockchain-based aid distribution framework using decentralized identities, smart contracts, and Zero-Knowledge Proofs for privacy-preserving eligibility verification.

CERTIFICATIONS

- IBM – Generative AI using watsonx
- VIT Summer Internship Program – Completion Certificate